Data Exploration-Assignment 1

Overview:

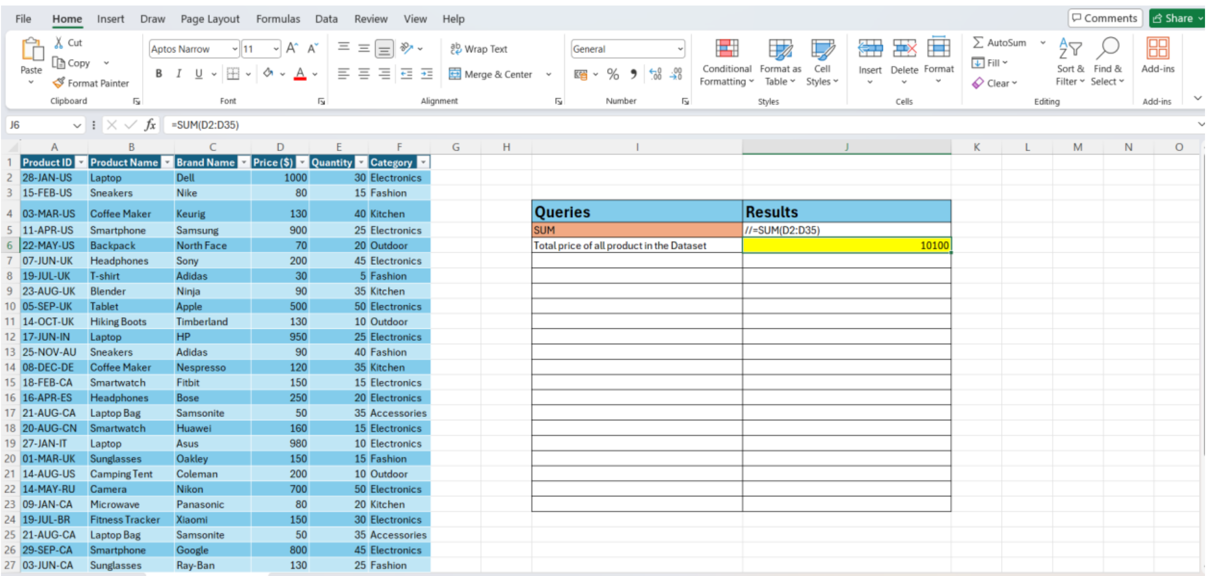
This given dataset is about a **Product Dataset for Data Analysis**.It contains 6 columns (product ID,Product Name,Brand Name,Price,Quantity,Category) and 35 rows.

Goal:

- Analyze product prices .
- Categorize them (High / Standard) .
- Perform calculations (SUM, AVERAGE, etc.) .
- Extract text from Product ID.

Screenshots:

1.SUM



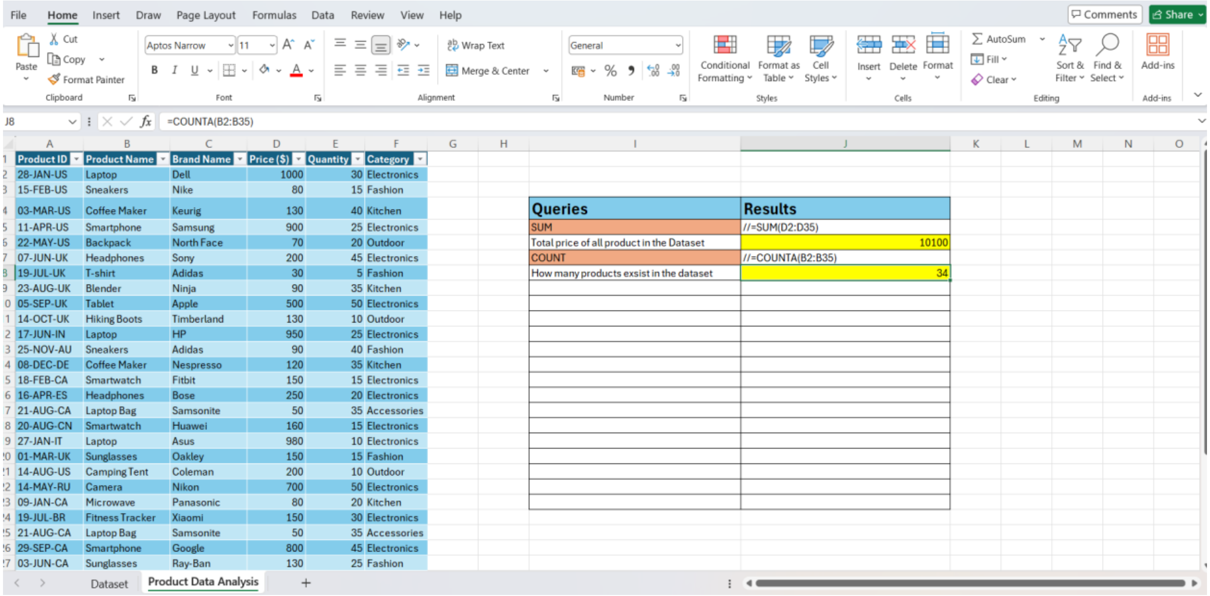
J6 =SUM(D2:D35)

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100

-The **SUM** function is the most efficient way to calculate total in Excel. It works on a range of cells at once.It avoids Manual Errors.

2.COUNT



=COUNTA(B2:B35)

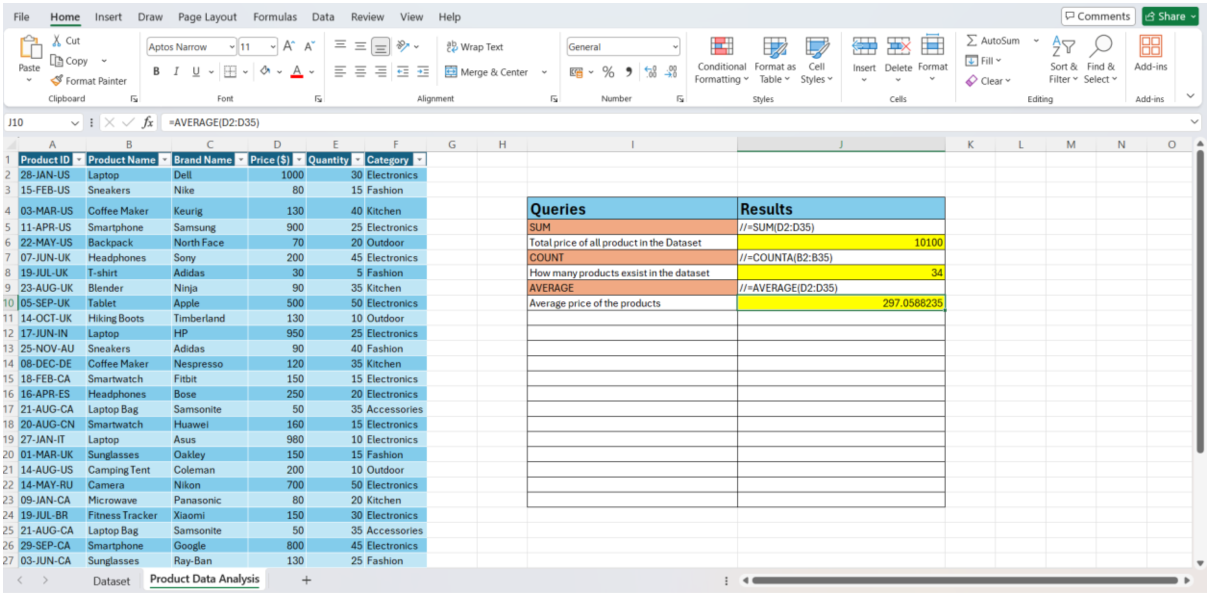
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34

Dataset | Product Data Analysis

-The Count function is used to count the no.of.cells.Here we use **COUNTA** because the product name is in text format.**COUNTA** is used to count the Non-Empty cells, where count is used to count cell having numeric values.

3.AVERAGE



=AVERAGE(D2:D35)

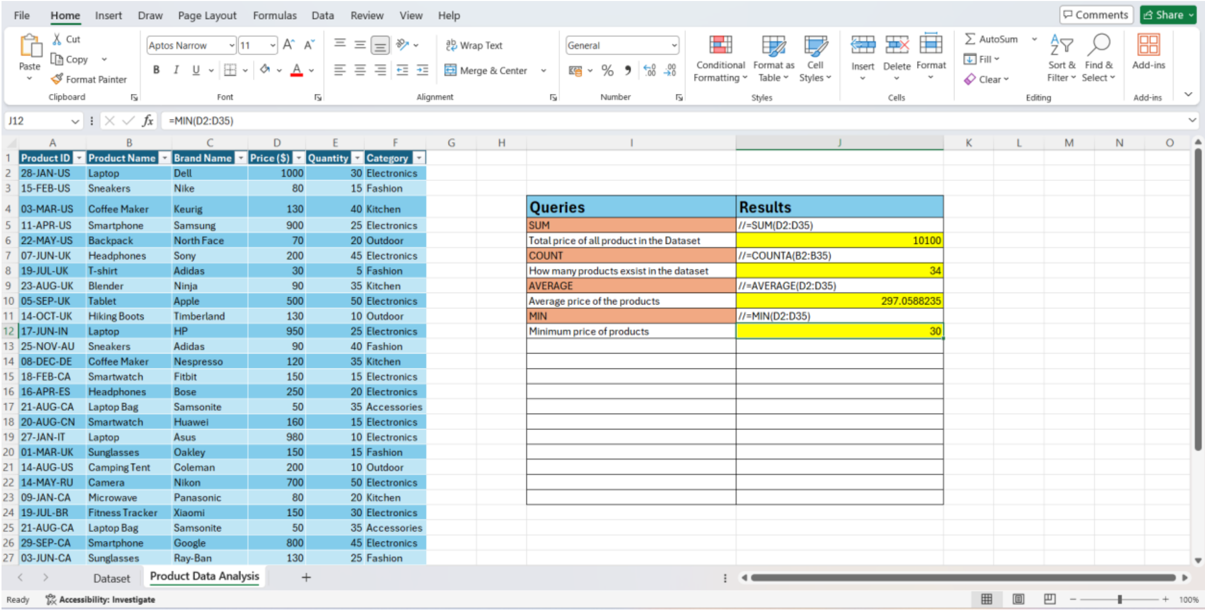
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34
AVERAGE	//=AVERAGE(D2:D35)
Average price of the products	297.0588235

Dataset | Product Data Analysis

-The **AVERAGE** function is used to find the mean (average) price of all products in the dataset. Instead of checking each value, we calculate one central value. This tells us the Average price level of products.

4.MIN



J12 =MIN(D2:D35)

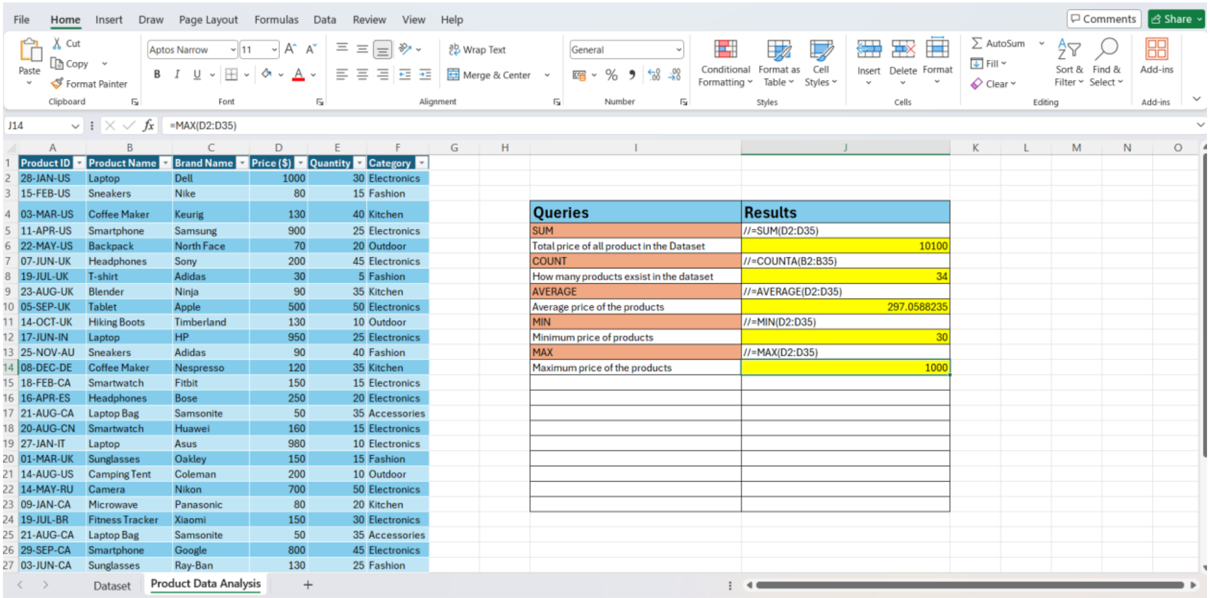
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34
AVERAGE	//=AVERAGE(D2:D35)
Average price of the products	297.0588235
MIN	//=MIN(D2:D35)
Minimum price of products	30

Dataset | Product Data Analysis

-The **MIN** function is used to identify the lowest price among all products in the dataset, helping to understand the minimum value in the data.

5.MAX



J14 =MAX(D2:D35)

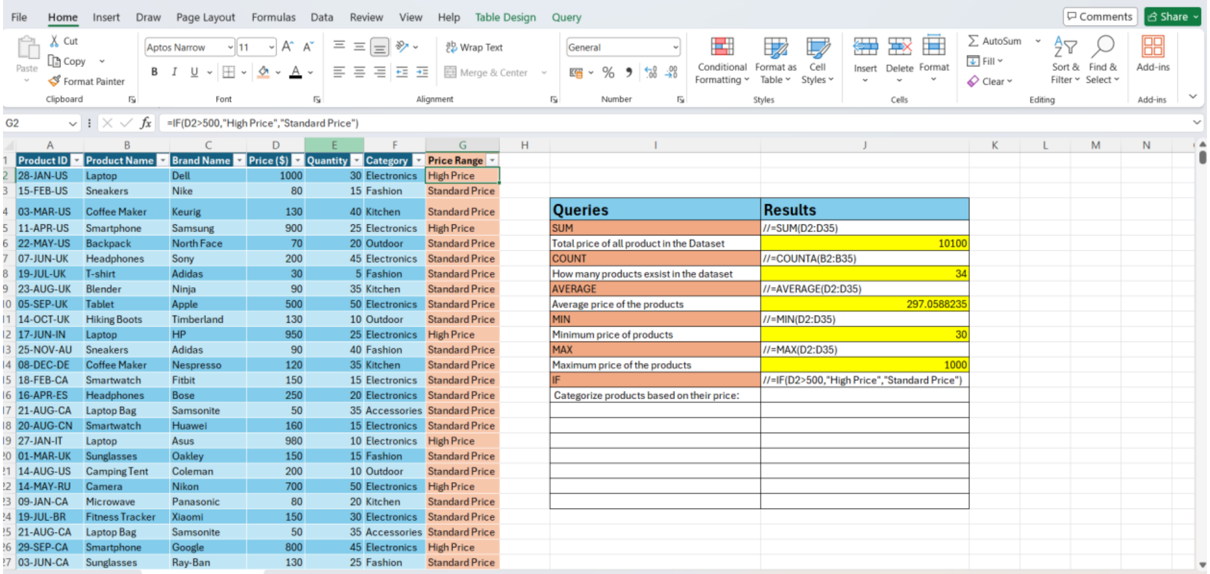
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34
AVERAGE	//=AVERAGE(D2:D35)
Average price of the products	297.0588235
MIN	//=MIN(D2:D35)
Minimum price of products	30
MAX	//=MAX(D2:D35)
Maximum price of the products	1000

Dataset | Product Data Analysis

- The **MAX** function is used to identify the largest price among all products in the dataset, helping to understand the maximum value in the data.

6.IF



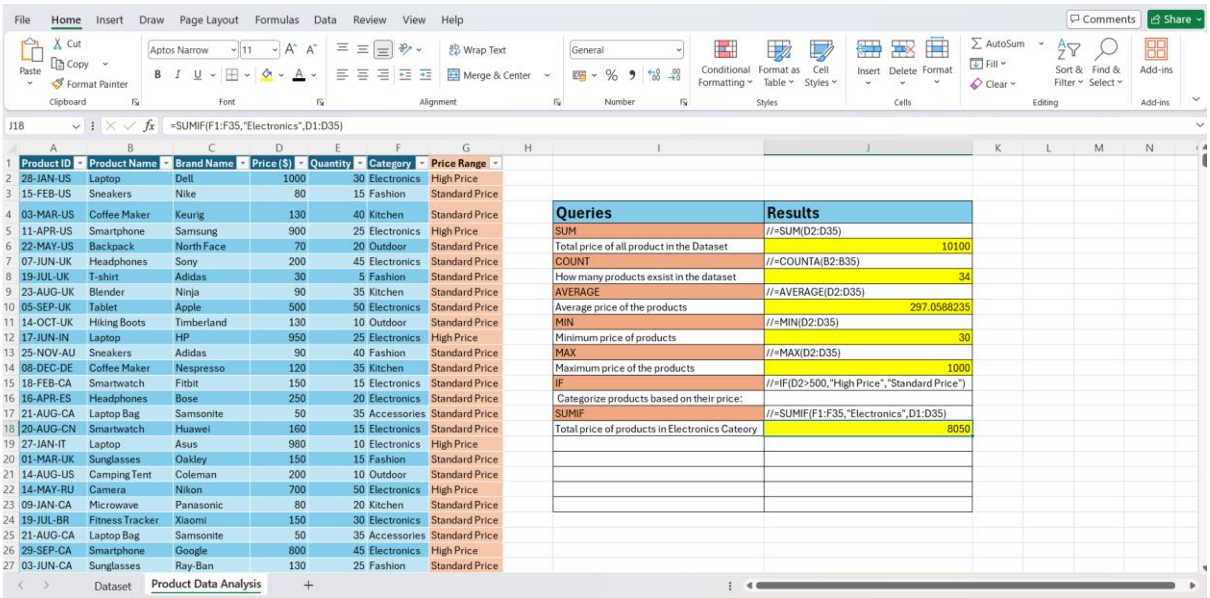
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price
05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard Price
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34
AVERAGE	//=AVERAGE(D2:D35)
Average price of the products	297.0588235
MIN	//=MIN(D2:D35)
Minimum price of products	30
MAX	//=MAX(D2:D35)
Maximum price of the products	1000
IF	//=IF(D2>500,"High Price","Standard Price")
Categorize products based on their price:	

-The **IF function** is used to apply logical conditions and categorize products into different price ranges based on their price values.

- IF checks a condition (D2 ≥ 500)
- If TRUE then it returns **High Price**
- If FALSE then it returns **Standard Price**

7.SUMIF

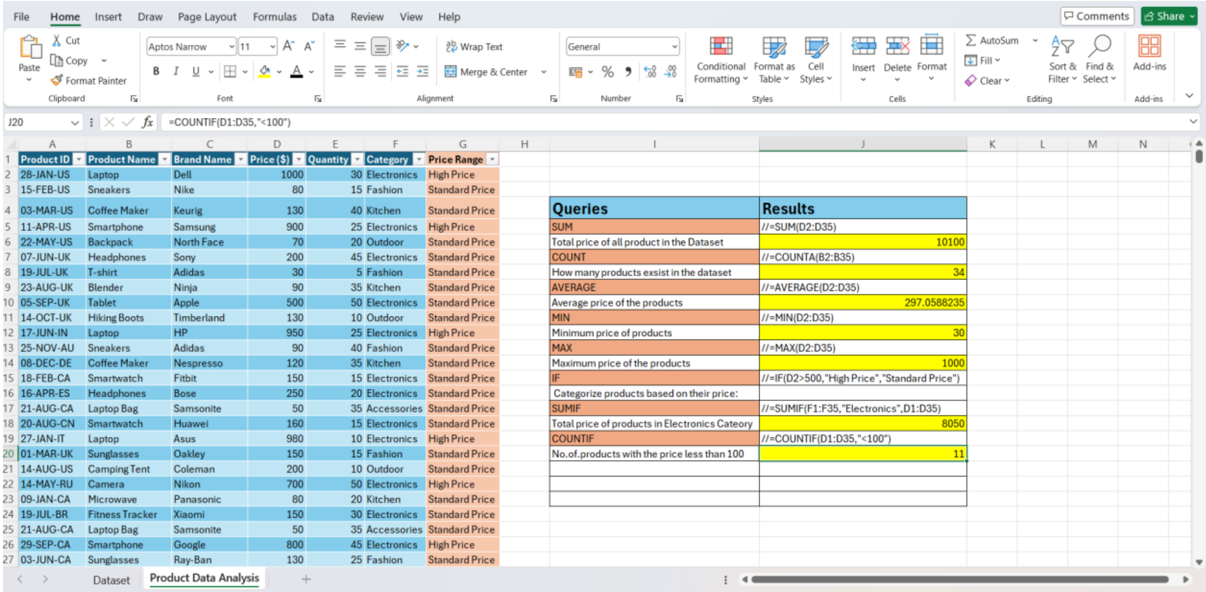


Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price
05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard Price
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price

Queries	Results
SUM	//=SUM(D2:D35)
Total price of all product in the Dataset	10100
COUNT	//=COUNTA(B2:B35)
How many products exsist in the dataset	34
AVERAGE	//=AVERAGE(D2:D35)
Average price of the products	297.0588235
MIN	//=MIN(D2:D35)
Minimum price of products	30
MAX	//=MAX(D2:D35)
Maximum price of the products	1000
IF	//=IF(D2>500,"High Price","Standard Price")
Categorize products based on their price:	
SUMIF	//=SUMIF(F1:F35,"Electronics",D1:D35)
Total price of products in Electronics Cateory	8050

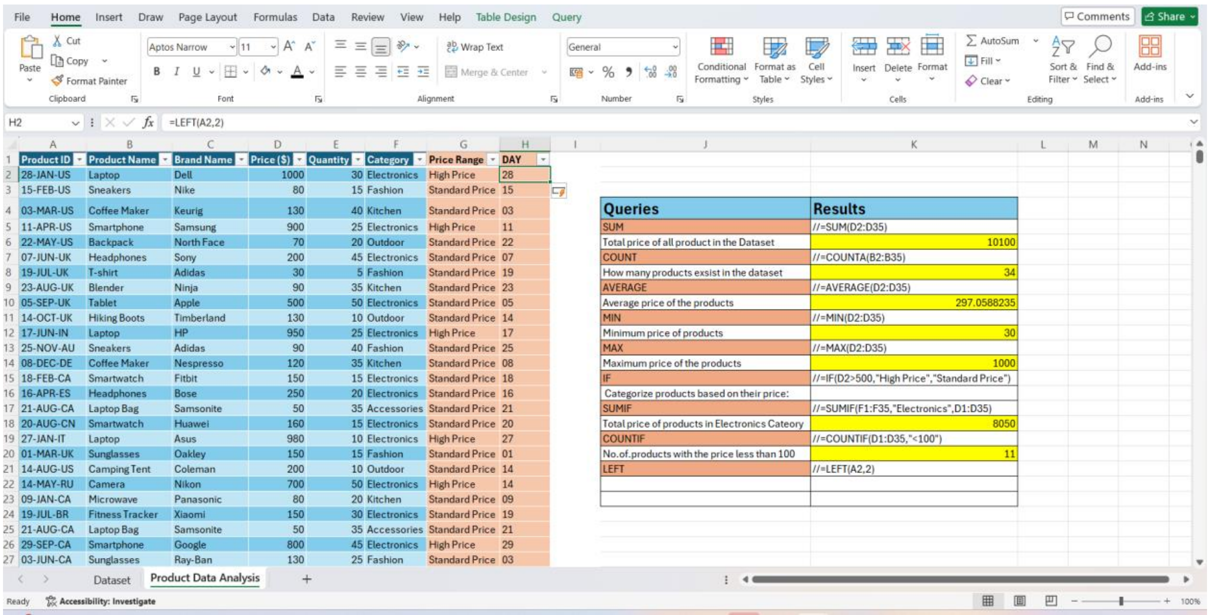
- The **SUMIF** function is used to add values based on a condition. In this dataset, it is used to calculate the total price of products in the Electronics category.

8.COUNTIF



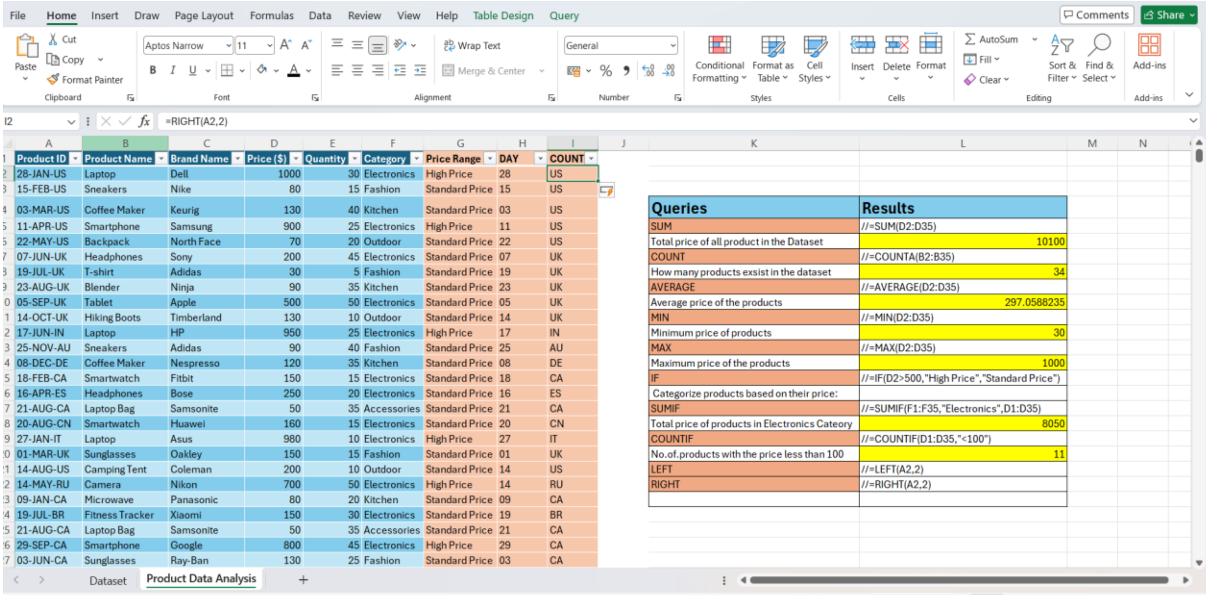
- The **COUNTIF** function is used to count the number of cells that meet a specific condition. In this dataset, it is used to count products with price less than 100.

9.LEFT



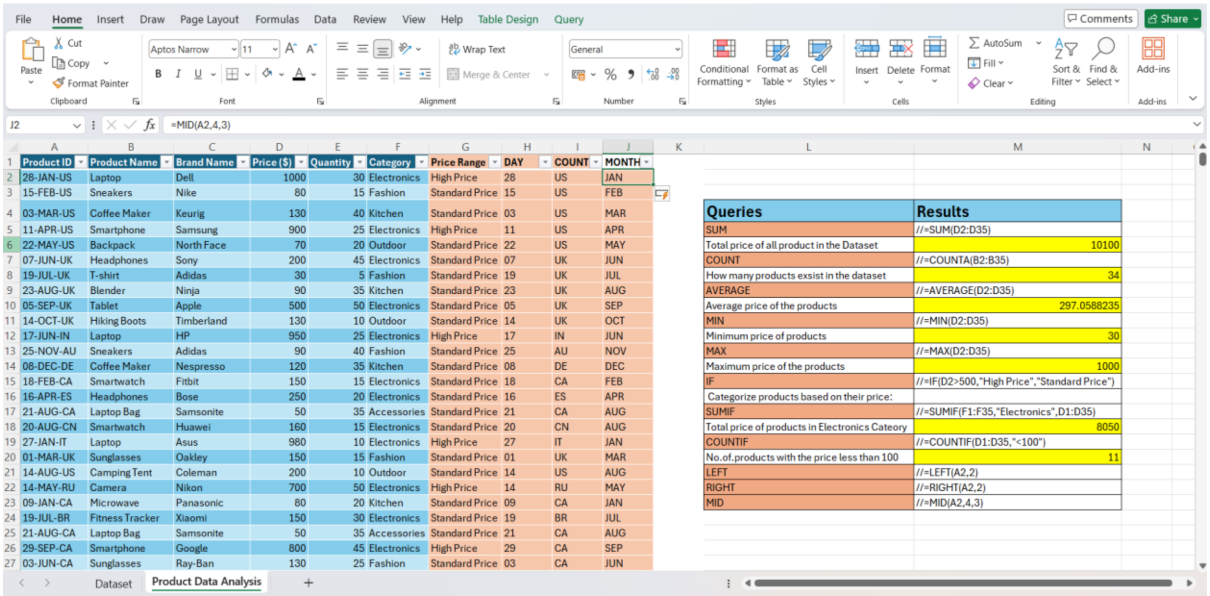
- The **LEFT** function is used to extract the first 2 characters from the Product ID.

10.RIGHT



- The **RIGHT** function is used to extract the last 2 characters from the Product ID.

11.MID



- The **MID** function is used to extract characters from the middle part of the product ID.